

Programmable Interval Timer (8253/8254)

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Course Title: Peripherals & Interfacing

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Lecture References:

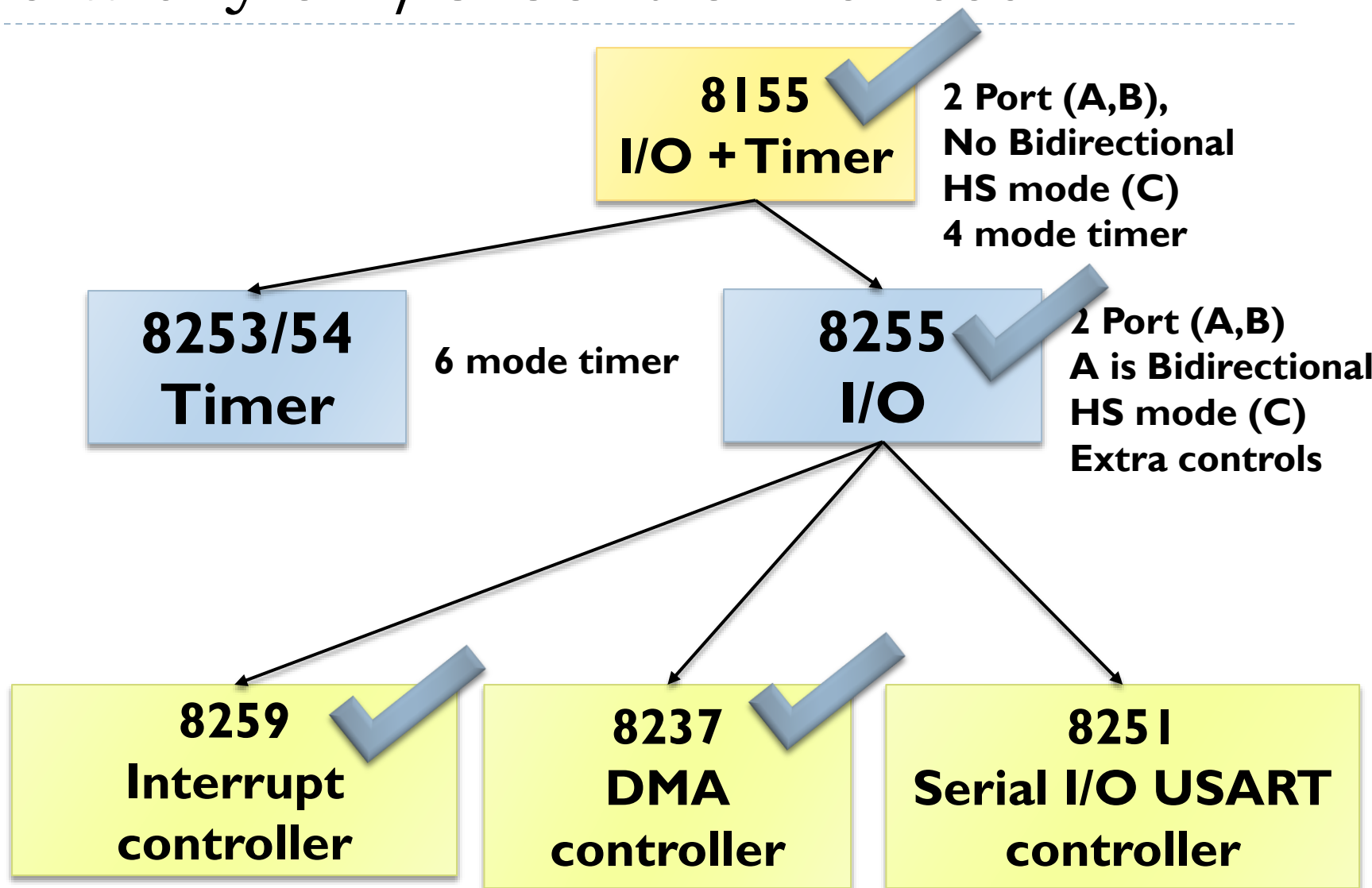
- ▶ Book:

- ▶ *Microprocessors and Interfacing: Programming and Hardware,*
Author: Douglas V. Hall

- ▶ Lecture Materials:

- ▶ *K.Vijay Kumar, Assistant Professor, Dept. of ECE*

Hierarchy of I/O Control Devices



Why Programmable Interval Timer

**Not
Possible To
Generate
Accurate
Time
Delays
Using Delay
Routines in
8086**



**Intel's Programmable
Counter/ Timer Device
(8253/8254) Facilitates**

- Accurate Time Delays
- Minimizes Load On Mp
- Real Time Clock
- Event Counter
- Digital One Shot
- Square Wave Generator
- Complex Waveform Generator

Why Does a Microprocessor Need a Timer Chip?

▶ **Timing is everywhere**

- ▶ Real-time clocks, delays, event counting, and waveform generation are needed constantly in embedded and general-purpose systems.

▶ **CPU time is expensive**

- ▶ Software delay loops waste processor cycles that could run other tasks — a dedicated chip does it for free, in parallel.

▶ **Precision & flexibility**

- ▶ A hardware counter driven by a stable clock gives microsecond-level accuracy that software loops cannot guarantee.

▶ **Therefore, the solution: 8253 / 8254**

8253 vs. 8254 PIT

8253

- **8253 can operate at frequency from dc to 2mhz**

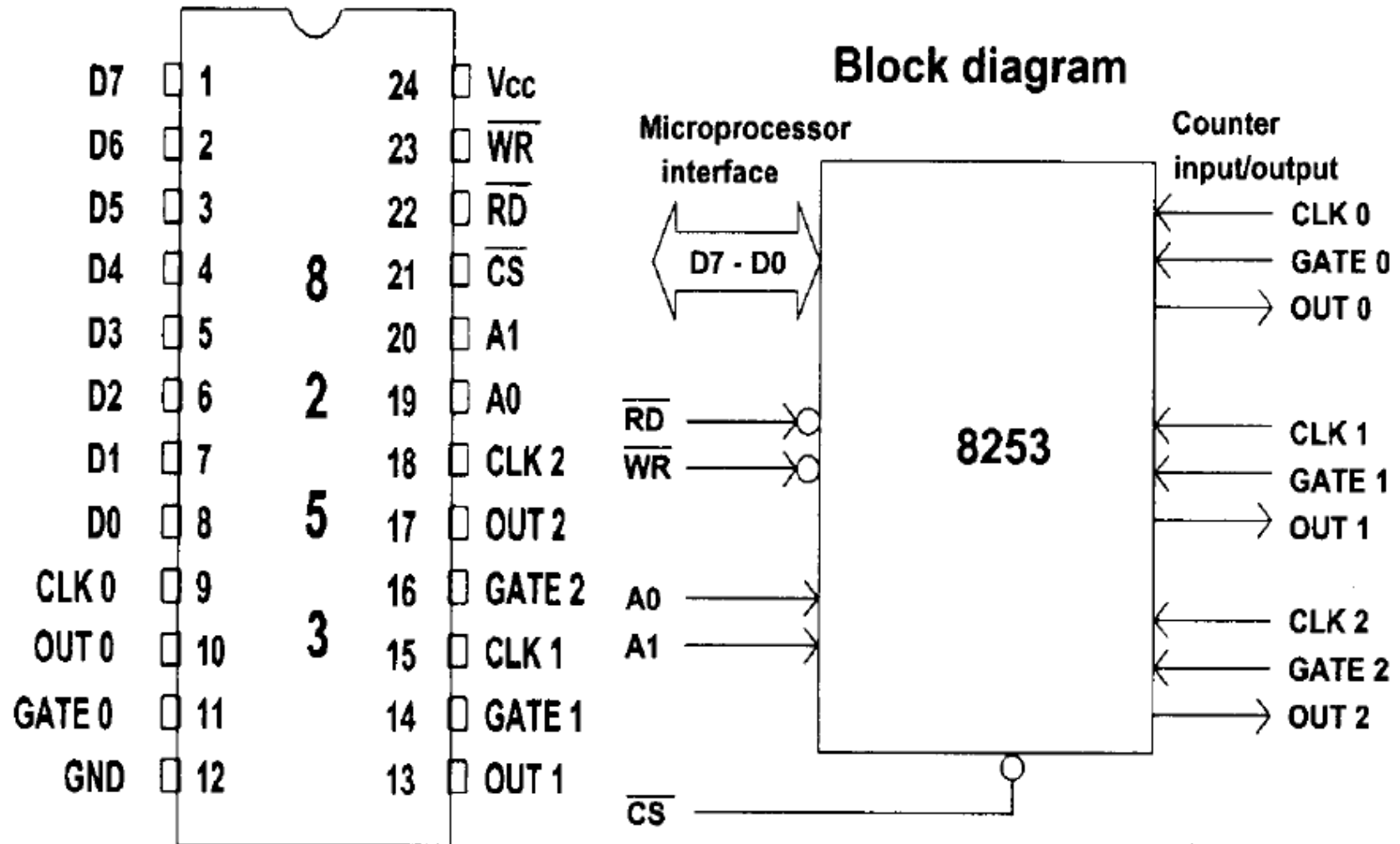
8254-ADVANCED VERSION OF 8253

- **8254 can operate with higher clock Frequency Range (DC To 8 Mhz AND 10 Mhz FOR 8254-2)**
- **Includes Status Read Back Command That Latches The Count And Status Of Counters**

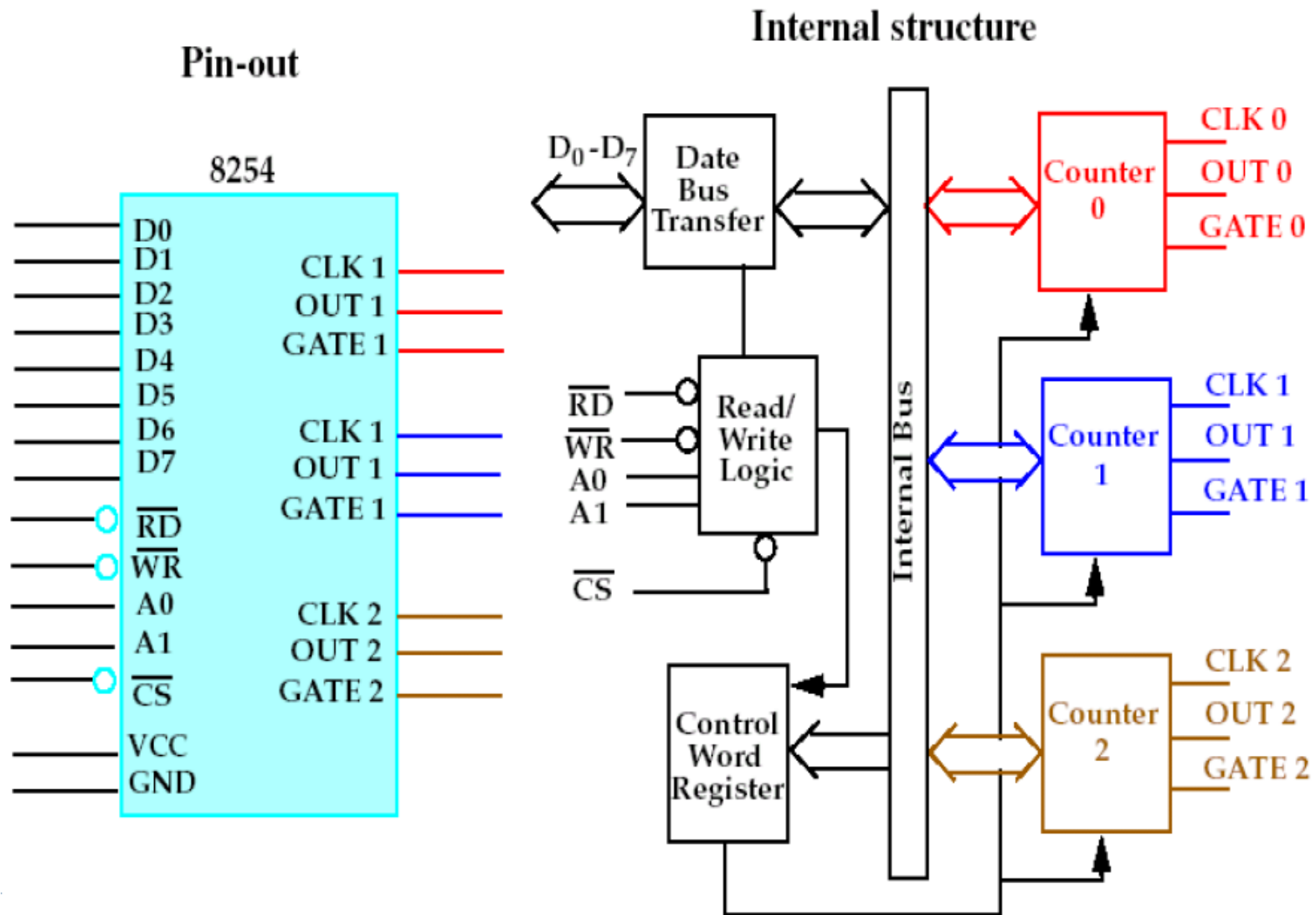
8253 vs. 8254 PIT

- ▶ Both compatible with All Intel and Most other Microprocessors and Handles Inputs up to 10 MHz
- ▶ The Intel 8254 is a counter/timer device designed to solve the common timing control problems in microcomputer system design.
- ▶ It provides **Three Independent 16-bit counters**, each capable of handling clock inputs up to 10 MHz
- ▶ Binary or BCD counting
- ▶ Single a +5V Supply and Standard Temperature Range
- ▶ **Six Programmable Counter Modes** and All modes are software programmable. The 8254 is a superset of the 8253.
- ▶ Used for controlling real-time events such as **real-time clock, events counter, and motor speed and data direction control.**

8253/8254 Pin and Block Diagram



8253/8254 Pin and Block Diagram



8253/8254 Pin and Block Diagram

▶ **Data Buffers:**

- ▶ 8 Bit Bidirectional D0-D7 Connected To Data Bus of Microprocessor
- ▶ In → Reads Data from Peripheral
- ▶ Out → Writes Data to Peripheral

▶ **Control Word Register:**

- ▶ Accepts 8 Bit Control Word Written by Microprocessor
- ▶ Can Only Be Written (**Read Not Possible !!**)
- ▶ Control Word Chooses **One of the Six Modes** of Operation

8253/8254 Pin and Block Diagram

▶ **Counters:**

- ▶ Three Counters – C1, C2 & C3
- ▶ Each 16 Bit Identical Presetable
- ▶ Down Counter Operates In BCD/Hex
- ▶ Controlled By Loading Count To Command Word Register
- ▶ “On The Fly” Reading is possible

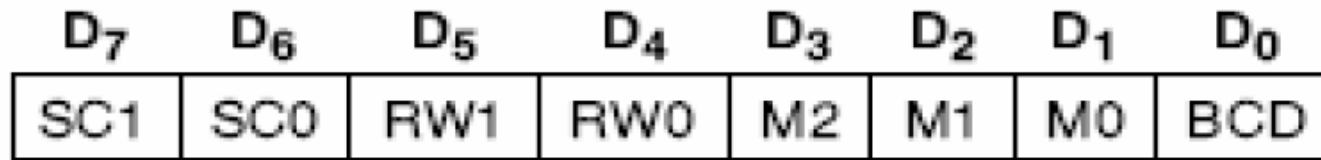
▶ **Control Logic:**

- ▶ CS – Logic 0 – Enables 8254
- ▶ RD – Logic 0 – Tells Microprocessor Reads Count from 8254
- ▶ WR – Logic 0 – Tells Microprocessor Writes Count/ Command into 8253/8254
- ▶ A1, A0 – Address Input Pins To Select Modes and Counters

8253/8254 Operation

$\overline{CS}$	$\overline{RD}$	$\overline{WR}$	A1	A0	OPERATION
0	1	0	0	0	Write Counter 0
0	1	0	0	1	Write Counter 1
0	1	0	1	0	Write Counter 2
0	1	0	1	1	Write Control Word
0	0	1	0	0	Read Counter 0
0	0	1	0	1	Read Counter 1
0	0	1	1	0	Read Counter 2
0	0	1	1	1	No Operation (Tristated)
0	1	1	X	X	No Operation (Tristated)
1	X	X	X	X	8254 Not Selected

Control Word Format



SC—Select Counter

SC1 SC0

0	0	Select Counter 0
0	1	Select Counter 1
1	0	Select Counter 2
1	1	Read-Back Command (see Read Operations)

RW—Read/Write

RW1 RW0

0	0	Counter Latch Command (see Read Operations)
0	1	Read/Write least significant byte only
1	0	Read/Write most significant byte only
1	1	Read/Write least significant byte first, then most significant byte

M—Mode

M2 M1 M0

0	0	0	Mode 0
0	0	1	Mode 1
X	1	0	Mode 2
X	1	1	Mode 3
1	0	0	Mode 4
1	0	1	Mode 5

BCD

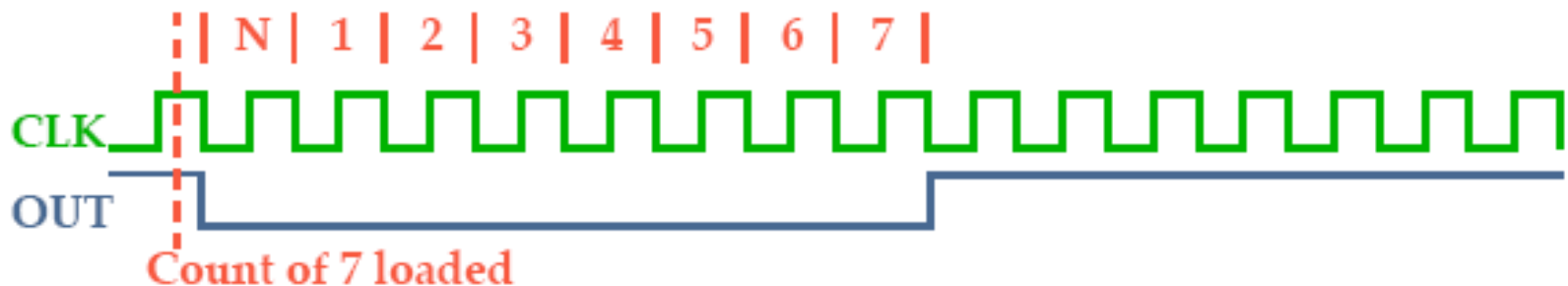
0	Binary Counter 16-bits
1	Binary Coded Decimal (BCD) Counter (4 Decades)

8254 Modes of Operation

- ▶ Mode 0 (Interrupt on Terminal Count)
- ▶ Mode 1 (Programmable Monoshot)
- ▶ Mode 2 (Rate Generator)
- ▶ Mode 3 (Square Wave Generator)
- ▶ Mode 4 (Software Triggered Strobe)
- ▶ Mode 5 (Hardware Triggered Strobe)

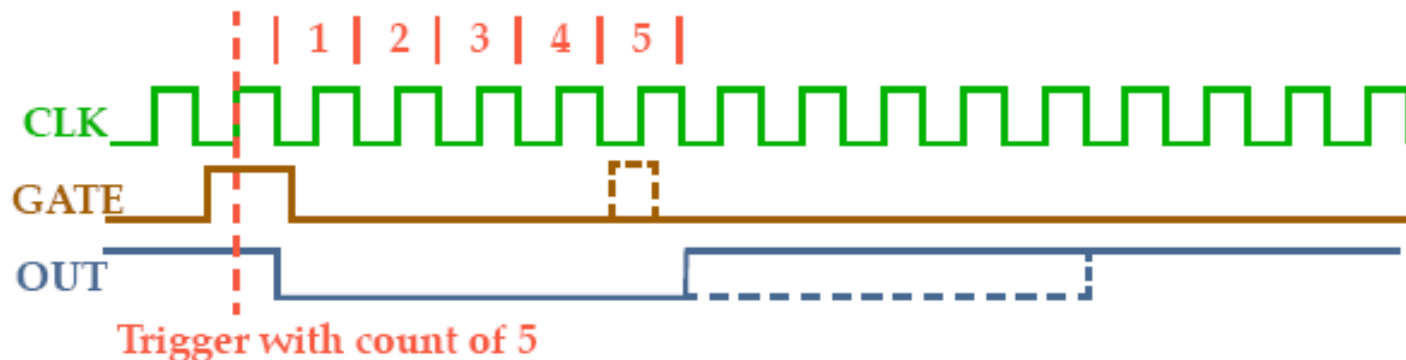
Mode 0: Interrupt on Terminal Count

- ▶ The OUT becomes a logic 0 when the control word is written remains low even after count value loaded in counter.
- ▶ Counter starts decrementing after falling edge of CLOCK
- ▶ The OUT goes high upon reaching the terminal count & remains high till reloading OUT can be used as interrupt
- ▶ Gate high → normal counting
- ▶ Gate low → counting terminated and current count latched till GATE goes high again



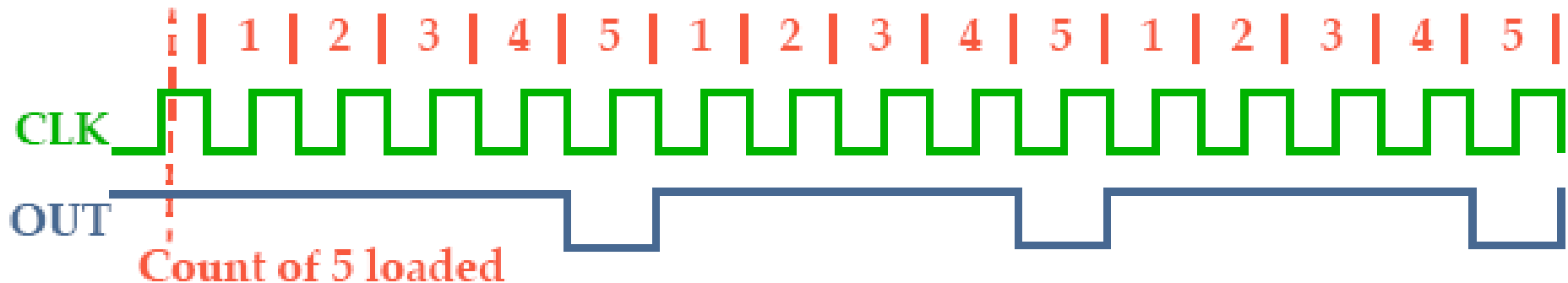
Mode 1: One-Shot Mode

- ▶ Monostable Multivibrator
- ▶ GATE as HIGH input is used as a trigger input
- ▶ OUT remains high till the count is loaded after application of trigger, OUT goes LOW and remains LOW till count becomes zero
- ▶ Another count loaded, when OUT is already LOW → it does not disturb counting **until a new trigger is applied at the gate**
- ▶ New counting starts after new trigger pulse



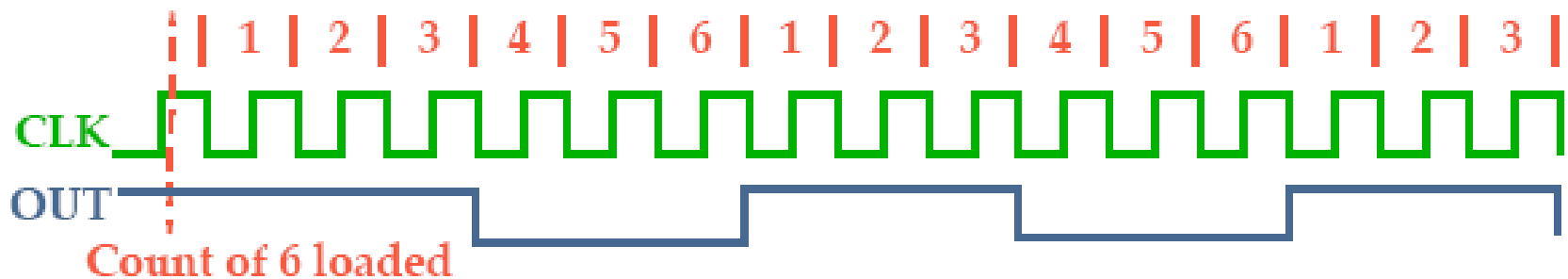
Mode 2: Rate Generator/Divide by N Counter

- ▶ When N is loaded as count → after N pulses → OUT goes low for only one clock cycle → then, count N is reloaded → OUT becomes high for N clock pulses
- ▶ The number of clock pulses between the two low pulses is equal to the count loaded
- ▶ Gate → logic 0 → no counting
- ▶ Gate → logic 1 → normal counting



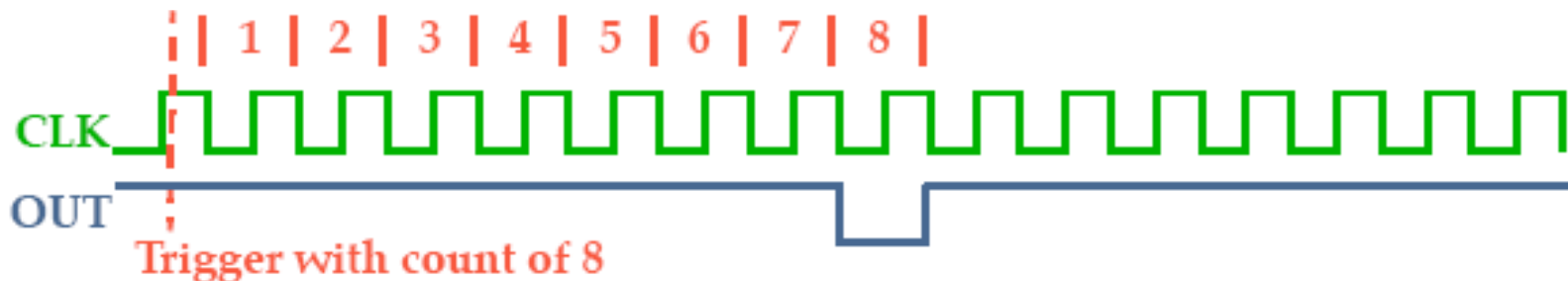
Mode 3: Square Wave Rate Generator

- ▶ When count N loaded is even $\rightarrow$ output remains HIGH for half the count and LOW for the rest half of the count
- ▶ When count N loaded is odd $\rightarrow$ output remains HIGH for $(N+1)/2$ and low for $(N-1)/2$.
- ▶ Repeated operation gives square wave
- ▶ Generates a continuous square-wave with G set to 1.
- ▶ If count is even, 50% duty cycle otherwise OUT is high 1 cycle longer.



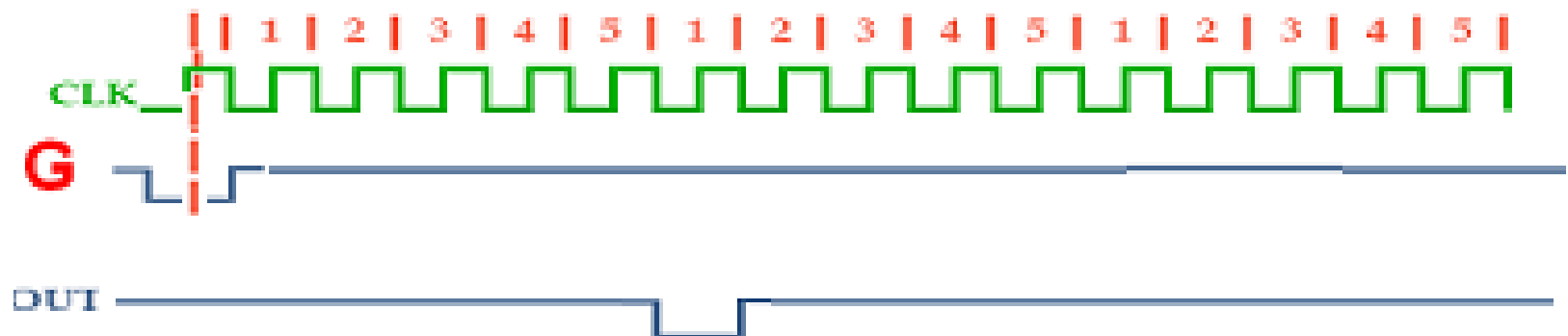
Mode 4: Software Triggered Strobe

- ▶ After mode is set output goes high
- ▶ When count is loaded counting down starts on reaching terminal count output goes low for only one clock cycle, and then again output goes HIGH
- ▶ The above said low pulse can be used as a strobe for interfacing MP with peripherals
- ▶ When GATE is LOW → counting is inhibited and count is latched. If a new count is loaded while counting, previous counting stops and new counting started in next clock cycle



Mode 5: Hardware Triggered Strobe

- ▶ This mode generates a strobe in response to the rising edge at the trigger
- ▶ Once mode is programmed and counter loaded, OUT goes HIGH
- ▶ Counter starts counting after the rising edge of the trigger (GATE)
- ▶ The OUTPUT goes LOW for one clock period, when the terminal count is reached
- ▶ Output will not go LOW until the counter content becomes zero after the rising edge of any trigger
- ▶ GATE is used as trigger input



Thank You !!

